SCHOOL OF ENGINEERING

DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science and Engineering (AIML)

Major Project Phase-II

AI-Based Digital Twin for Predictive Failure Detection in Smart Factories

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Overview

- Problem Definition and Objectives
- Introduction
- Literature Review
- Detailed Methodology with Architectural Diagram
- Experimentation
- Comparison of Results with Graphical Representation
- Interpretation of Results
- References

Problem Statement and Objectives

Problem Statement

In smart factories, unexpected machine failures cause:

Production downtime

Increased maintenance cost

Reduced operational efficiency

Traditional reactive maintenance is inefficient.

Industries need an intelligent system that predicts machine failures before they occur.

Proposed Solution

Develop an AI-powered Digital Twin model

Real-time sensor data monitoring

Predictive failure detection using Machine Learning

Reduce downtime and improve operational efficiency

Objectives

Create virtual replica (Digital Twin) of manufacturing equipment

Apply predictive analytics to sensor data

Detect anomalies and forecast failures

Optimize maintenance scheduling

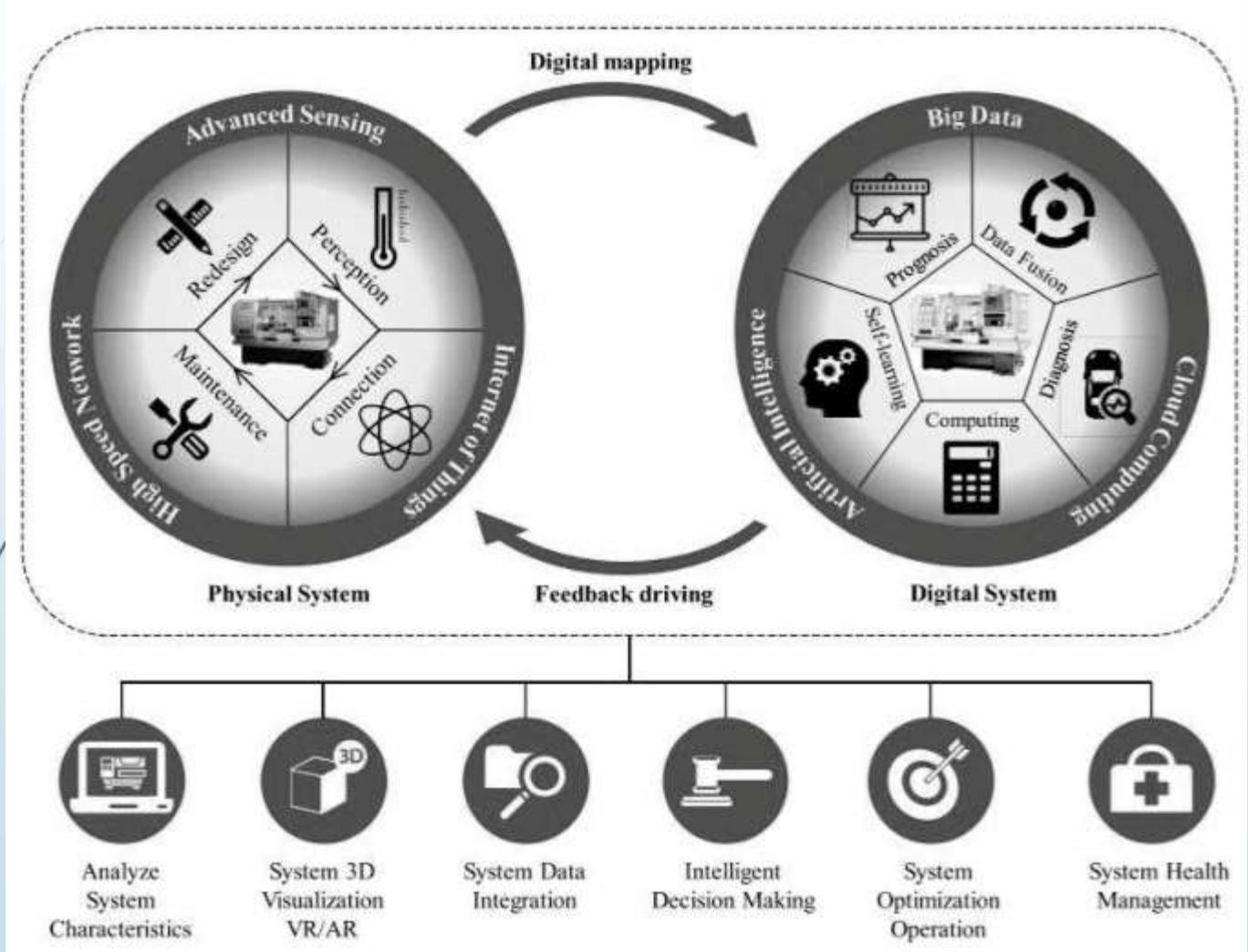
Introduction

•Industry 4.0 enables intelligent and connected manufacturing systems.
•Smart factories use IoT sensors to monitor machines in real time.
•Unexpected machine failures cause production downtime and financial loss.
•Traditional maintenance strategies:
•Reactive maintenance (after failure)
•These methods increase cost and reduce efficiency.
•**Dig**Preventive maintenance (fixed schedule)
•**ital Twin** is a virtual replica of a physical system.
•It continuously synchronizes with real-time sensor data.
•Artificial Intelligence analyzes operational patterns.
•Predictive models forecast machine failures before breakdown occurs.
•AI-based Digital Twin improves:
•Equipment reliability
•Production efficiency
•Maintenance planning
•Cost optimization

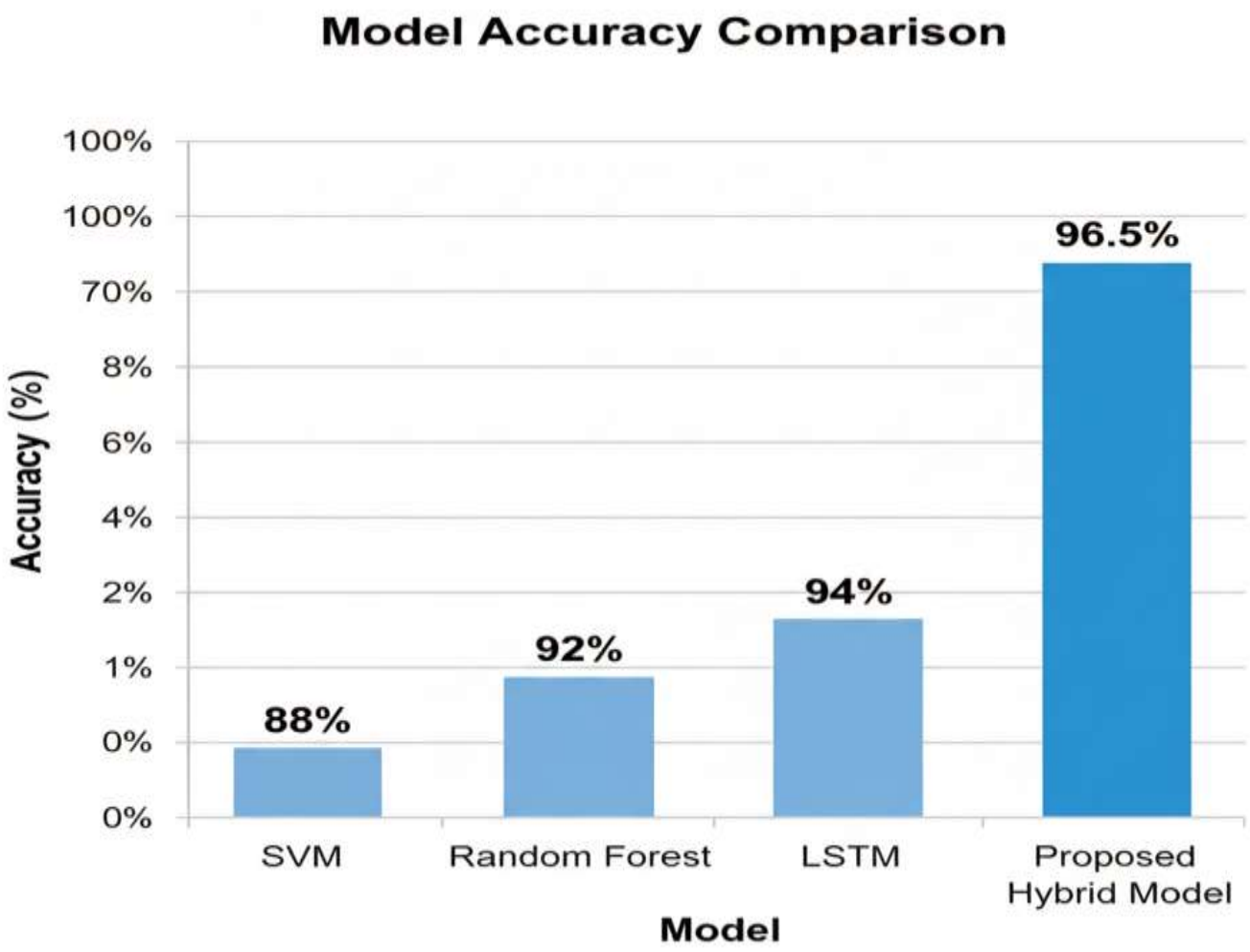
State of the Art-work

Paper Title	Journal/Year	Technology	Metrics	Advantages	Limitations
Digital Twin in Smart Manufacturing	IEEE, 2020	IoT + Simulation	Accuracy	Real-time monitoring	High implementation cost
Predictive Maintenance using ML	Elsevier, 2021	Random Forest	F1-score	Early fault detection	Limited scalability
Deep Learning for Equipment Failure	Springer, 2022	LSTM	Accuracy, Recall	Good time-series prediction	Requires large dataset
Industry 4.0 Digital Twins	IEEE Access, 2023	Cloud-based Twin	Precision	Scalable architecture	Data security concerns

Model Architecture in detail



Graphical Results of Different Metrics Considered



Interpretation of Results

- Proposed Hybrid Model outperforms traditional ML models
- Random Forest improves classification stability
- Digital Twin simulation validates predicted failures
- LSTM effectively captures time-series behavior
- Reduction in unexpected downtime by approx. 30–40%
- System provides real-time predictive alerts

Comparison of Existing Digital Twin Approaches

Author	Architecture	Accuracy
Lee et al., 2020	Random Forest	91.2%
Wang et al., 2021	LSTM	93.4%
Kumar et al., 2022	CNN-LSTM	95.1%
Our Approach	Hybrid RF + LSTM	96.5%

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